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Carbon starvation enhances microbiologically influenced corrosion in marine offshore infrastructures

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Introduction: In offshore wind foundations, microbiologically influenced corrosion (MIC) is a critical concern in the mud zone, where steel is embedded in sediment characterized by limited oxygen availability and pronounced redox gradients. In marine sediments, the limited availability of labile substrates can influence microbial metabolic strategies, potentially promoting alternative electron-transfer pathways. Here, we performed an exploratory mechanistic study to investigate how carbon availability influences microbial community structure and corrosion behavior in multispecies biofilms relevant to offshore infrastructure.

Methods: Sediment from the North Sea was used as inoculum, and steel coupons were exposed under blank (no added carbon), lactate, and yeast extract conditions. Corrosion was assessed by weight loss and 3D profiling, while microbial communities were analyzed using 16S rRNA gene sequencing.

Results and discussion: The findings illustrated that nutrient regime strongly shaped microbial and chemical conditions. Yeast extract produced the highest total sulfide concentrations (9 mM), but corrosion severity did not scale with bulk sulfide accumulation. The highest corrosion rate occurred under carbon-limited conditions (0.55 mm yr^{-1}), compared with 0.122 and 0.072 mm yr^{-1} for lactate and yeast extract, respectively. Pitting rate was also most pronounced under carbon-limited conditions (0.65 mm yr^{-1}). These findings suggest that the test conditions for offshore foundation materials should include low-labile-carbon scenarios to better reflect sediment zones and highlight the need for further mechanistic investigation.

KEYWORDS
carbon starvation, microbiologically influenced corrosion (MIC), offshore wind foundation, pitting corrosion, sulfate-reducing bacteria

1 Introduction

Marine corrosion remains one of the main degradation processes affecting metallic materials exposed to seawater, driven by the coupled effects of salinity, dissolved oxygen, moisture, hydrodynamics, and biological activity (Kotu et al., 2025). Corrosion severity varies across marine exposure zones (including atmospheric, splash, tidal, submerged, and

mud zones) each characterized by distinct physicochemical and microbiological conditions that govern corrosion mechanisms and damage morphology. Among marine infrastructures, offshore wind structures have emerged as critical assets in global decarbonization strategies. Their structural reliability depends fundamentally on the long-term durability of steel foundations operating under aggressive mechanical and environmental loading.

Offshore wind foundations (such as monopiles) experience cyclic loading from wind, waves, and rotor operation, resulting in significant fatigue stresses that frequently control structural design life. Superimposed on these mechanical loads, corrosion [particularly microbiologically influenced corrosion (MIC)] causes material loss, localized corrosion (such as pitting), and stress concentration, thereby accelerating fatigue crack initiation and propagation (Shojai et al., 2025). The most critical loading region frequently occurs just below the seabed within the so-called mud zone, where bending moments are highest (Momber, 2024). Yet this region remains among the least accessible sections of offshore foundations for inspection and maintenance. As a result, uncertainties in corrosion rates, mechanisms, and spatial variability within sediments are commonly addressed through conservative design assumptions, increasing costs while still leaving residual risk.

The mud zone represents a uniquely complex corrosion environment. Steel in this region is in direct contact with marine sediment, where oxygen penetration is limited, creating pronounced redox gradients. Such conditions favor anaerobic microbial activity and MIC. Cathodic protection may also be less effective in sediments due to reduced ionic conductivity and shielding effects, further increasing susceptibility to localized corrosion (Momber, 2024). Consequently, understanding the MIC mechanisms in near seabed region is essential for realistic service-life prediction of offshore monopiles. MIC encompasses a spectrum of processes by which microorganisms modify corrosion behavior, either indirectly through metabolic activity called metabolite-mediated MIC (M-MIC) or directly through electrochemical interactions with metal surfaces entitled extracellular electron transfer (EET)-mediated MIC (EET-MIC) (Xu et al., 2023; Tüccar et al., 2025). In natural multispecies biofilms, these mechanisms rarely operate independently; rather, they occur simultaneously within spatially structured communities where redox stratification, metabolite retention, and microbe–surface interactions collectively govern corrosion processes (Xu et al., 2023). Importantly, MIC mechanisms arise from the interplay between microbial community structure and a broad range of environmental factors, with carbon availability being one (though not the only) parameter influencing metabolic pathways and electron-transfer processes (Yuxuan et al., 2022).

Traditionally, MIC has primarily been interpreted as a metabolite-driven process, in which microorganisms utilize available organic substrates and generate corrosive metabolic products such as hydrogen sulfide, organic acids, or differential aeration conditions that accelerate metal degradation. Within this framework, environments with limited bioavailable organic carbon are often assumed to pose a lower MIC threat because reduced nutrient availability is expected to constrain microbial activity and metabolite production. This assumption is particularly relevant for marine sediments surrounding offshore foundations, where a large fraction of the organic matter is refractory and only a limited portion is readily accessible to microorganisms (Jørgensen and Marshall, 2016). However, growing evidence indicates that low nutrient availability does not necessarily correspond to low corrosion severity. Pure-culture laboratory studies have demonstrated that some microorganisms can shift toward utilizing metallic iron or other energetic metals as alternative electron donors under carbon-limited conditions, potentially accelerating corrosion through EET-associated processes (Xu and Gu, 2014; Cui et al., 2025; Li et al., 2021). For example, carbon-starved *Desulfovibrio vulgaris* biofilms have been shown to significantly accelerate nickel corrosion by shifting toward metal-dependent electron acquisition pathways (Pu et al., 2023). These findings suggest that carbon limitation can fundamentally alter microbial survival strategies and intensify microbe–metal interactions. However, it remains insufficiently understood whether low bioavailable carbon in marine sediments hosting complex natural microbial consortia necessarily results in lower MIC threat for offshore wind foundations. Addressing this question is particularly important for offshore wind foundations in the mud zone, where limited nutrient availability, restricted mass transport, and complex biofilm development may strongly influence localized corrosion processes and long-term structural degradation.

In this study, sediment samples collected from the North Sea were used as inoculum to investigate the role of carbon availability in shaping microbial community composition and corrosion behavior. Three conditions were established: (i) unamended sediment representing baseline carbon-limited conditions, (ii) amendment with lactate as a defined low-molecular-weight electron donor, and (iii) amendment with yeast extract as a complex organic substrate. By coupling corrosion rate measurements with microbial community analysis, we evaluated how carbon limitation and nutrient amendment influenced sediment microbial structure and corrosion response. The results provide insight into potential starvation-associated MIC processes in near-bed marine environments and contribute to a better understanding of MIC threats affecting offshore monopile foundations. This approach allowed us to assess treatment-dependent patterns in corrosion, sulfide production, and microbial community structure, while providing a basis for future studies to examine the specific corrosion mechanisms in more detail.

2 Materials and methods

2.1 Materials

Seawater and sediment samples were collected from the North Sea by Endures BV Company (the Netherlands). The seawater samples were obtained near the Endures facility, close to the shoreline (Nieuwe haven, Den Helder, the Netherlands), while the sediment samples were collected from a site near an offshore wind park at approximately 5 m below the seafloor surface. Sediment samples were collected using a hollow liner to retrieve vertical sediment cores. The cores were sealed and transported under anaerobic conditions. Endures subsequently subsampled

the material and supplied 1 L of sediment to our laboratory in airtight bottles under a nitrogen headspace to maintain anoxic conditions. The exact *in situ* temperature during sampling was not recorded; however, sampling took place in September, when typical North Sea temperatures range between approximately 15–20 °C. Prior to experimental use, samples were stored at 4 °C. All laboratory experiments were conducted under a fume hood, and the temperature was 21 ± 0.5 °C, which is within the upper range of natural environmental conditions. The seawater used in this study was characterized by a chloride concentration of 15,824 mg L^{-1}, nitrate of 1.96 mg L^{-1}, and sulfate of 2,135.8 mg L^{-1}. Electrical conductivity was 4,306 µS cm^{-1}, and the initial pH was 8.26.

Carbon steel coupons were used as metal specimens in this study (0.453 ± 0.02 g). According to manufacturer specifications (Goodfellow GmbH, Hamburg, Germany), the nominal chemical composition was: Fe 99.5%, Mn 0.3%, Si 0.1%, C < 0.08%, P < 0.04%, and S < 0.05%. The material was supplied as larger sheets and machined into coupons with dimensions of 0.8 × 0.8 × 0.1 cm. While the chemical composition of the coupons represents a relatively pure low-carbon steel, offshore wind monopiles are typically manufactured from structural steels (e.g., S355) containing higher levels of alloying elements such as Mn and trace microalloying additions. These differences may influence mechanical properties and corrosion behavior; however, the selected material provides a simplified and reproducible model system to investigate fundamental corrosion and MIC processes.

2.2 Experimental set-up

Prior to use, the seawater was autoclaved at 120 °C for 20 min, whereas the sediment sample was used as a natural microbial inoculum. This approach ensured controlled and reproducible initial conditions, with seawater primarily acting as the electrolyte phase and sediment providing the microbial consortia driving MIC processes. Prior to exposure, carbon steel coupons were cleaned by acid pickling in 2% HCl, followed by neutralization in 1.2 M NaOH. The coupons were then thoroughly rinsed with deionized water, rinsed with acetone to remove residual moisture, and dried under a stream of nitrogen gas (SP0775-2013, 2013). The cleaned coupons were then weighed using a precision analytical balance.

The experimental system (Figure 1) consisted of a closed-loop setup (using MINIPULS 3 Peristaltic Pumps, Gilson, USA) containing 5 mL of sediment as an inoculum, placed at the bottom of each column and 1 L of circulating sterilized seawater. Three nutrient treatments were investigated (yeast extract, sodium lactate, and no nutrient). In each column, three coupons were suspended in the overlying water column (Figure 1). Coupons were spatially separated using sterile glass beads (Ø = 0.238 cm) to prevent contact. Prior to assembly, columns and glass beads were autoclaved at 120 °C for 20 min, and all tubing was sterilized using 70% (v/v) ethanol. Anoxic conditions were established by purging the entire system with nitrogen gas prior to the start of the experiment. The setups were maintained under a fume hood at 21 ± 0.5 °C. The initial pH was 8.26. Initially, columns were inoculated and incubated under static conditions for 3 days to allow initial biofilm establishment, after which the system was operated at a flow rate of 0.05 mL min^{-1} and maintained for 25 days. The flow rate of 0.05 mL min^{-1} was selected to represent a low-flow, weakly mixed condition rather than turbulent bulk seawater flow. In offshore environments, hydrodynamic conditions near steel surfaces are highly heterogeneous. While exposed monopile surfaces may experience wave- and current-driven turbulent flow, regions near the seabed, within sediment-contact zones, and beneath biofilm or corrosion-product deposits are often governed by weaker shear and boundary-layer-controlled mass transport. Under such conditions, substrate delivery, metabolite retention, and biofilm development can differ substantially from well-mixed bulk flow environments, which is particularly relevant for MIC processes (Christiansen et al., 2026; Rivera-Alvarez et al., 2020).

Yeast extract (1 g L^{-1}) and sodium lactate (3.5 g L^{-1}) were added separately as organic carbon amendments. Each biotic treatment was prepared in triplicate, together with sterile abiotic controls. Sodium lactate was selected as a defined low-molecular-weight organic electron donor commonly used in sulfate-reducing bacteria (SRB) media and MIC studies, including formulations derived from Postgate media (Stein et al., 2025). Yeast extract was selected as a complex nutrient supplement frequently used to stimulate microbial growth and biofilm development in laboratory cultivation systems. While lactate represents a simple carbon source for microbial growth. The concentrations were chosen based on commonly applied ranges in MIC-related studies and were intended to compare the effects of simple vs. complex carbon amendments. Therefore, three treatments were considered: Blank/no nutrient, Lactate, and Yeast conditions.

2.3 Corrosion layer/biofilm analysis

After the experimental period, carbon steel coupons were carefully removed from the system for morphological and compositional analyses. To preserve the integrity of the biofilm and corrosion products, separate sets of coupons were used for biofilm/corrosion layer thickness measurements and for elemental analysis and focused ion beam–scanning electron microscopy (FIB–SEM), although all coupons originated from the same loop/bottle within each treatment. This approach minimized disturbance, oxidation, or alteration of the biofilm during repeated handling and preparation.

For biofilm and corrosion layer thickness analysis, coupons were gently washed once with filter-sterilized PBS/dH_2O (1:2, v/v), then incubated overnight at 4 °C in 1 mL 2.5% (v/v) glutaraldehyde solution for fixation of microorganisms. Subsequently, the coupons were washed first with filter sterilized PBS/dH_2O (1:2, v/v) for 5 min, then for another 5 min in sterilized dH_2O. The dehydration of coupons was carried out using different ethanol dilutions (30% [30 min], 50% [30 min], 70% [30 min], 80% [60 min], 90% [60 min] (v/v) and absolute [60 min]) and dried with N_2 gas (An et al., 2020). Surface morphology and corrosion layer thickness were then quantified using a 3D optical profilometer VR-3000 (Keyence Corporation, Osaka, Japan). Surfaces were analyzed by white-light interferometry and focus-variation techniques to generate